A Lossless Secret Image Sharing Scheme based on Bit Sharing Visual Cryptography

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Abstract — **Visual Cryptography is a technique to share the secret information in the images for achieving data/information security. The technique uses encryption and decryption techniques for protecting the data. The secret image is encrypted into number of shares and distributed to the multiple participants. In decryption, the secret image can be reconstructed by combining the subset of the shares using simple computations. In the proposed secret image sharing technique, secret image is partitioned or divided into four shares based on the bit sharing approach. These shares are then covered with different cover images before distributing to the participants. All four shares are required to reconstruct the original secret image.**

Keywords— Secret Image; Cover Image; Shares; Encryption; Decryption;

I. INTRODUCTION

Nowadays most of the information is shared between users through internet where it is more prone to the attacks and hacking. Therefore, it is more important to secure the information while sharing or transmitting through internet. Visual Cryptography (VC) is technique for achieving security on visual information such images, text and videos. In Visual Cryptography, the image information is encrypted/encoded and generate the multiple share or shadow images. In decoding or decryption process the original image is recovered by human visual system (HVS). Decryption process does not require any computational device, and it is a mechanical operation where all shares are stacked together to observe the reconstructed image using human visual system (HVS) without any complex cryptographic calculations and computations. Thus, it eliminates the drawback of hardware and software requirement, which is needed for the decryption process in traditional cryptography [1]. In addition to dividing secret image into multiple shares, the shares are embedded in or combined with the chosen cover images. Then, Visual cryptography allows the encryption of secret information in the image form and that adds a meaningful cover image to each share.

In the secret sharing based on visual cryptography, secret image and cover images are inputted to the share synthesizer to encrypt the secret. The secret is divided into multiple shares and each share is combined with cover image. Shares are outputted from the share synthesizer and distributed to the multiple participants as shown in the figure1. In the decryption, the original secret image is reconstructed from the received shares using extraction and reconstruction method as shown in the figure1. The technique uses very simple computation for reconstruction of the original secret image.

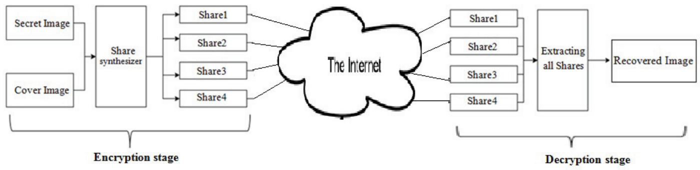

Fig1. Model of Security Sharing based on Visual Cryptograph

The Paper is organized as follows. Section 2 describes the related works. In Section 3, the proposed method is introduced and described with an example. In section 4, Experimental results along with performance of the proposed method is presented. Finally, the Section 5, gives the conclusion.

II. RELATED WORKS

Blakley [2] and Shamir [3] was introduced the first idea of secret sharing scheme autonomously. Later numerous efficient secret sharing techniques proposed and appeared in light [4, 5, 6, 16] based on polynomial based scheme, threshold based schemes and Huffman coding schemes. Another compose of secret sharing scheme called visual cryptography (VC) is proposed by Naor at el [1].In a similar line numerous different systems of visual cryptography techniques were proposed based on Latin squares [7], based on blue-noise dithering principles[8], based on optimal pixel expansion[9], based on embedded random shares [10], based on Position Exchange Technique [11], based on bit slicing technique[12] and so on. Visual Cryptography is not restricted to gray-level images and binary images; it can be helpful for encryption of color images as well. Later some more Visual Cryptography systems for color image encryption were proposed [13, 14, 15]. In this paper, we have proposed bit sharing techniques to create the share images from the secret image and simple bitwise OR operations to combine the shares with cover images.

III. PROPOSED METHOD

In the proposed method, Gray scale Secret image can be securely transmitted using bit sharing based visual cryptography technique. Since the secret image is assumed to

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be gray scale image, each pixel is represented as set of 8 bits as shown in the figure2.

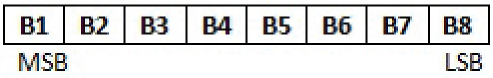

Fig2. Representation of Pixel - Gray scale image

Secret image is encrypted by splitting secret image into multiple shares and each share is combined with cover image. Each pixel of secret image is processed and its 8 bits are shared among four shares. Initially, all pixels of all four shares are considered be black pixels i.e. its intensity is 0. For each pixel of the secret image, the first two bits of the pixel (B1 and B2) are embedded in the last two LSB positions of the corresponding pixel of the Share1; next two bits of the pixel (B3 and B4) are embedded in the last two LSB positions of the corresponding pixel of the Share2; next two bits of the pixel (B5 and B6) are embedded in the last two LSB positions of the corresponding pixel of the Share3; and last two bits of the pixel (B7 and B8) are embedded in the last two LSB positions of the corresponding pixel of the Share4. Four different gray scale cover images are chosen, all which are of same size of secret image. In each cover image, the last two bits (two LSB positions) of each pixel are set to 0. The secret shares and cover images are combined by OR operation as shown in the figure3.

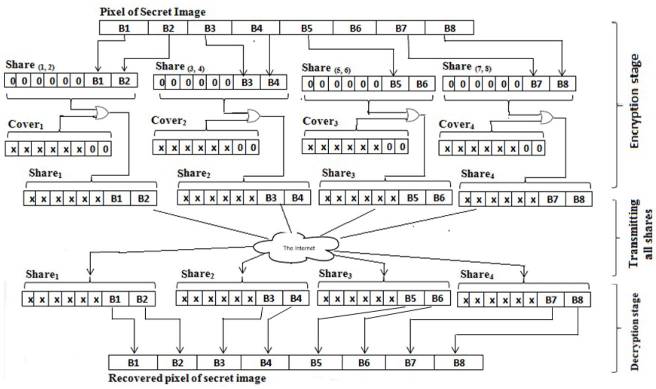

Fig3. Proposed Model of Bit Sharing based Visual Cryptography

A. Encryption Algorithm

```
Input: secret[m][n];
       cover1[m][n];    cover2[m][n];
       cover3[m][n];    cover4[m][n];
Output:share1[m][n];    share2[m][n];
       share3[m][n];    share4[m][n];

/* initialization for shares */
for i = 0 to m-1
  for j = 0 to n-1
    share(1, 2)[i][j]=0;        share(3, 4)[i][j]=0;
    share(5, 6)[i][j]=0;        share(7, 8)[i][j]=0;
  end // j for loop
end // i for loop

/* preprocess cover images */
for i = 0 to m-1
  for j = 0 to n-1
    cover1[i][j].pixel[7]=0;    cover1[i][j].pixel[8]=0;
    cover2[i][j].pixel[7]=0;    cover2[i][j].pixel[8]=0;
    cover3[i][j].pixel[7]=0;    cover3[i][j].pixel[8]=0;
    cover4[i][j].pixel[7]=0;    cover4[i][j].pixel[8]=0;
  end  // j for loop
end // i for loop

/* combining share images and cover images */
for i = 0 to m-1
  for j = 0 to n-1
    share1[i][j]=share(1, 2)[i][j] OR cover1[i][j];
    share2[i][j]=share(3, 4)[i][j] OR cover2[i][j];
    share3[i][j]=share(5, 6)[i][j] OR cover3[i][j];
    share4[i][j]=share(7, 8)[i][j] OR cover4[i][j];
  end  // j for loop
end // i for loop

/*pixel bit sharing of secret image */
for i = 0 to m-1
  for j = 0 to n-1
    share(1, 2)[i][j]→pixel[7] = secret[i][j]→pixel[1];
    share(1,2)[i][j]→pixel[8] = secret[i][j]→pixel[2];
    share(3, 4)[i][j]→pixel[7] = secret[i][j]→pixel[3];
    share(3,4)[i][j]→pixel[8] = secret[i][j]→pixel[4];
    share(5, 6)[i][j]→pixel[7] = secret[i][j]→pixel[5];
    share(5, 6)[i][j]→pixel[8] = secret[i][j]→pixel[6];
    share(7, 8)[i][j]→pixel[7] = secret[i][j]→pixel[7];
    share(7, 8)[i][j]→pixel[8] = secret[i][j]→pixel[8];
end  // j for loop
end // i for loop
```

B. An example

For example, the first pixel intensity value of the secret image is assumed to be 147. The pixel representation is given as follows: (1 0 0 1 0 0 1 1)

Since, secret image is splitting into four shares, technique uses four different cover images. And, the first pixel values of each cover images (cover1, cover2, cover3 and cover4) are assumed to be 175, 94, 230 and 35 respectively.

The encryption and decryption of the first pixel of the secret image is demonstrated. The process and resulted shares of the encryption and decryption are presented in the following example as shown in the figure 4 and figure 5 respectively.

C. Security

- Since the secret image pixel bits are shared among the shares in the two LSB positions of the respective pixels of cover image, cover image appearance (very minute change in the pixel intensity) is same as cover image chosen before combining with secret shares.

- All shares are required to reconstruct the original secret image. Someone with less than four shares cannot retrieve any information of original secret (not possible to reconstruct the original secret with less than four shares).

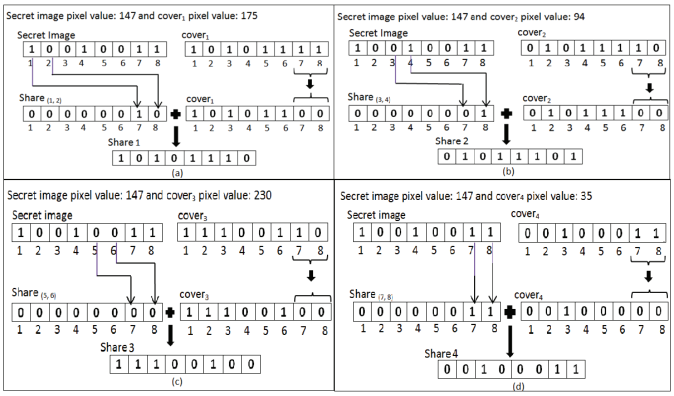

Fig4. An example: Encryption of a pixel of the secret image
a) pixel computed for share1 b) pixel computed for share2
c) pixel computed for share3 d) pixel computed for share4

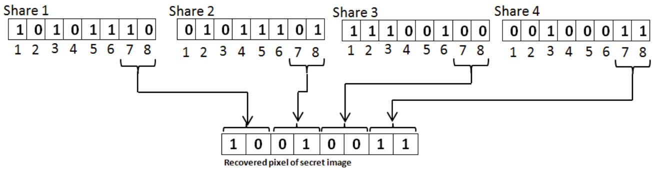

Fig5. An example: Decryption of a pixel

IV. Implementation and Results

Proposed approach uses visual cryptography on grey scale images with bit sharing technique. It is implemented to encrypt the grey scale images and experimental results are presented in the fig 6. Secret image (fig 6 (a)) is partitioned or divided into four shares based on the bit sharing technique and combined with chosen cover image (fig 6 (b)). After encryption, the computed four shares (fig 6 (c-f)) are distributed to the participants. In the decryption, recovered secret image which is computed from the received all four shares is presented in the fig. 6 (g). The results shown that recovered secret image is lossless and its resolution is same as that of original secret image. image. Performance time for encryption and decryption functions of the proposed work are computed and presented in the table1.

Table1. Observations of Proposed Model Performance in terms of time required for encryption and decryption of Secret Image of 260 X 346

OPERATION		***TIME***
Encryption	Generating four shares	188ms
	combining shares with cover image	159ms
Decryption	reconstruction of secret image	266ms

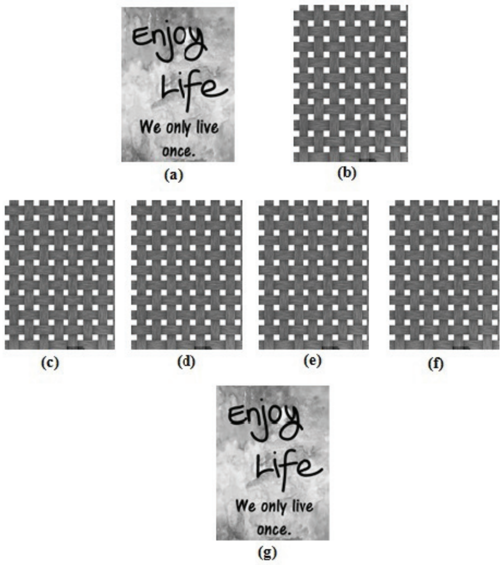

Fig6. a) Secret image b) Cover image c-f) Share 1 to 4
g) Recovered secret image

V. Conclusion

A bit sharing based visual cryptography method on grey scale images is proposed in this work. Secret image is encrypted into four shares based on the bit positions and combined with cover images. Secret image is recovered from the shares without any loss of data during decryption. The performance of the proposed method is explored in terms of its execution time. The execution time for encryption and decryption is very feasible since it is involved with all simple operations.

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